BHARGAV RAIKAR

Chemical Engineer (B.E. Chemical, 2027) | National Award Winner, SIH 2025

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PROFESSIONAL SUMMARY

I'm Bhargav Raikar, a Chemical Engineering graduate-in-progress at UICT Jalgaon, specializing in process design and industrial innovation. I bring hands-on experience in PID development, equipment design, process simulation, and plant optimization, backed by practical engineering projects. As a National Award Winner of Smart India Hackathon 2025 (AIR 1) and a CHEMATHON 2026 finalist, I transform complex engineering challenges into practical, sustainable solutions. My goal is to contribute my technical expertise, problem-solving mindset, and innovation-driven approach to designing safer, more efficient, and sustainable industrial processes. (*Atmanirbhar Bharat*).

SKILLS

Process Design & Documentation: PFD, P&ID, Heat & Material Balance, Equipment Sizing & Datasheets, Hydraulic & Line Sizing, Control Philosophy, MOC Selection, Process Optimization.

Core Chemical Engineering: Mass & Energy Balance, Heat & Mass Transfer, Fluid Mechanics, Thermodynamics, Reaction Engineering, Distillation, XRD, FTIR, Research And Development.

Equipment & Unit Operations: Reactors, Heat Exchangers, Pumps & Piping, Filtration, Drying & Calcination, Heat Recovery, Reverse Osmosis

Plant Operations & Control: PLC , SCADA Fundamental PID Control, Control Loops & Interlocks, Start-up / Yield & Energy Optimization, Troubleshooting

Process Safety & HSE: HAZOP, Permit-to-Work & Lockout-Tagout Awareness, PSV & Relief Basics, Root Cause Analysis, Near-Miss Reporting, ETP & Air Pollution Control, Water Recovery, MPCB Compliance

Software & Tools: Aspen Plus, DWSIM, MATLAB, AutoCAD, MS Excel, Web-based Process Calculators

Professional Skills: Team Leadership, Project Management, Technical Documentation, Problem Solving, Presentation & Communication

INTERNSHIP & PROJECTS

Chemical Engineering Intern | JNK Chemdist Process Solutions 2026

- Gained exposure to Process, Project, Sales and R&D activities, including industrial process workflows and equipment used in chemical plants.
- Developed a basic understanding of Aspen Plus process simulation, PFDs, P&IDs and preliminary process line-sizing concepts.
- Studied distillation-column internals and packing types, and gained exposure to reactors, heat exchangers, storage tanks and evaporators.
- Prepared a basic cost sheet, supported routing of a reactor enquiry to Sales, and contributed to an R&D concept for a Mountain-Water-Inspired Water Treatment System.

N-PHANTOM: Multifunctional Stealth Coating for Defence (AIR 1) Oct 2025 – Present

Smart India Hackathon 2025, ***National Award Winner (Technical Head)*** *[Project Link]* UICT, Jalgaon

- Engineered an economical (21.6%) multi-signature coating system, applicable as a retrofit, that simultaneously reduces Radar Cross-Section (RCS), infrared (IR) emissions and thermal signatures for defence platforms.
- Integrated thermally stable acrylic/siloxane binders with IR-reflective (92%) pigments and radar-absorbing ferrites for broadband EM attenuation, with corrosion resistance and field durability, supporting India's defence self-sufficiency.

Modular Single-Reactor Hydrothermal System (Waste-to-Oil) Jan 2026 – Mar 2026

CHEMATHON 2026 IIT PATNA Project (Team Lead) *[Project Link]* UICT, Jalgaon

- Designed a modular HTL reactor converting unsorted municipal solid waste (food scraps, textiles, plastics) into liquid hydrocarbons, eliminating costly feedstock pre-sorting.
- Achieved 80–95% feedstock conversion and 70–85% liquid yield through optimised temperature, pressure and residence-time control.
- Integrated recuperative heat exchangers recovering 65–75% of process energy, making the system self-sustaining at pilot scale.

EDUCATION

.University Institute of Chemical Technology (UICT). (2023 – 2027)

Bachelor of Engineering in Chemical Engineering Jalgaon, Maharashtra, India

- CGPA: 6.02 / 10

Shatabdi Science College, Agaskhind (2022 – 2023)

12 th HSC Science Nashik, Maharashtra, India

- Percentage: 55.33% / 100%

CERTIFICATIONS & ACHIEVEMENTS

- **Smart India Hackathon 2025: National Award Winner (AIR 1)** *National Level Innovation Award*
- **CHEMATHON 2026 Finalist** | | **Chemical Process Safety Certification**
- **Chemical Process Safety Certification** | Industrial Safety & Regulatory Compliance.
- **National Roller Skating competition** | First Place.